BLOCKCHAIN

TECHNOLOGY

Project Presentation

2025 / 2026

Ethereum Sepolia Testnet

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Developing a Blockchain-Based Solution for

Women Empowerment

in Semi-Urban Rwanda

Application Area

Supply Chain Management

Smart Contract

WomenSupplyChain.sol

Network

Ethereum Sepolia Testnet

Tests

23 / 23 Passing ✓

```
Contract: 0xAaB5646895d0F81dc45C0D6D8c3f9807db9F96D9
```

Agenda

01 **Problem Identification**
The challenge facing women-led businesses in semi-urban Rwanda

02 **Solution Design**
How blockchain technology addresses the problem

03 **Smart Contract**
WomenSupplyChain.sol — architecture and key functions

04 **Implementation**
Deployment to Ethereum Sepolia Testnet

05 **Testing & Validation**
23 automated test cases — all passing

06 **Impact & Conclusion**
Social impact and future development roadmap

01 Problem Identification

Counterfeit Products

Authentic handmade goods from women cooperatives are undercut by cheap imitations. Buyers cannot verify product origin or authenticity.

No Market Trust

Buyers have no way to verify where a product came from, who made it, or how it was processed — creating information asymmetry against producers.

Financial Exclusion

Without auditable production records, women-led businesses cannot access credit, government programs, or export markets.

68% of women SMEs lose market share to counterfeits | < 12% maintain auditable records | $700M+ annual financing gap

02 Solution Design

How Blockchain is Utilized

- Immutability: Product records cannot be altered once written
- Decentralization: No single authority controls the platform
- Smart Contracts: Payments auto-execute when conditions are met
- Transparency: All stages queryable by anyone on Etherscan
- Wallet Identity: No bank account needed — just MetaMask

5 Key Features

1. Business Registration — wallet-linked business profile
2. Product Listing — name, category, price on-chain
3. Supply Chain Tracking — per-stage journey logging
4. Direct ETH Purchase — instant payment to producer
5. Availability Control — owner manages listing status

03 Smart Contract — WomenSupplyChain.sol

Business	owner · name · location isRegistered · registeredAt
Product	id · name · category · producer price · isAvailable · stageCount
SupplyStage	stageName · location · notes recordedBy · timestamp
Purchase	productId · buyer · seller price · timestamp

Access Control Modifiers

```
onlyRegisteredBusiness()
onlyProductOwner(productId)
productExists(productId)
```

Blockchain Events

```
emit BusinessRegistered(…)
emit ProductRegistered(…)
emit StageAdded(…)
emit ProductPurchased(…)
```

04 Implementation & Deployment

1

npm init + Install Hardhat v2.22.0

2

Write WomenSupplyChain.sol in VS Code

3

Configure hardhat.config.js + .env

4

npx hardhat compile → Success

5

Get 0.05 Sepolia ETH from Google Faucet

6

npx hardhat run scripts/deploy.js --network sepolia

```
✓  Deployed: 0xAaB5646895d0F81dc45C0D6D8c3f9807db9F96D9  |  Deployer: 0xfacaA0D21...C42  |  Balance: 0.05 ETH
```

05 Validation & Testing

23

PASSING

Business Registration TC-01 to TC-05	5
Product Registration TC-06 to TC-10	5
Supply Chain Tracking TC-11 to TC-15	5
Product Purchase TC-16 to TC-20	5
Product Availability TC-21 to TC-23	3

```
$ npx hardhat test

  TC-01 ...   TC-23  →  23 passing (52s)

WomenSupplyChain deployed to: 0xAaB5646895d0F81dc45C0D6D8c3f9807db9F96D9
```

06 Social Impact & Conclusion

500K+
Women SME owners in semi-urban Rwanda

30–50%
Middleman value captured (eliminated)

$700M+
Annual financing gap addressed

0
Banks needed to participate

Mission Alignment — Technology as an Equalizer

- Democratizes trust — any buyer can verify product authenticity without trusting a single institution
- Removes intermediaries — ETH transfers go directly from buyer to producer wallet in real time
- Builds digital credit history — on-chain transactions serve as proof of business activity for microfinance
- No bank required — participation needs only a smartphone with MetaMask wallet

Thank You

Blockchain-Based Supply Chain for Women Empowerment in Semi-Urban Rwanda

Contract: 0xAaB5646895d0F81dc45C0D6D8c3f9807db9F96D9

Sepolia Testnet | Solidity 0.8.19 | Hardhat | 23/23 Tests Passing